

Empirical Analysis of Denoised Data Construction in Single Image Denoising

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Abstract

*Image denoising is a foundational task in computational photography and image processing, yet obtaining clean ground-truth images is often impractical in real-world (e.g. low-light, microscopy, or long-exposure imaging). Supervised denoising methods deliver strong performance but rely on large volumes of clean-noisy paired data. Self-supervised paradigms such as Noise2Noise [3] and Noise2Void [2] remove the need for clean references, and single-image variants such as DIP [7], Self2Self [6] and R2R [5] further relax the data requirement to a single noisy observation. However, all of these approaches rely on information-lossy operations – masking, downsampling, dropout, or re-corruption – that discard useful image content and bound the achievable denoising quality from above. In this report, we review these works and analyze **Positive2Negative (P2N)** [4], a CVPR 2025 paradigm that performs self-supervised denoising from a single noisy image. Leveraging the approximately zero-mean and symmetric nature of real noise, P2N constructs positive and negative noisy images by re-noising the predicted clean image with the predicted noise, and supervises the network through a denoised consistency loss. Our code and datasets are available at https://github.com/bscny/IP_Final_Project.*

1. Introduction

Image denoising plays a vital role in enhancing visual quality and serves as a foundational technique across a wide range of computational photography applications, including low-light imaging, mobile photography, and fluorescence microscopy. Existing methods for image denoising can be broadly categorized into two major paradigms based on their data requirements: methods relying on multi-image observations and single-image-based methods.

Supervised denoising networks, such as those trained on the SIDD dataset [1], rely on pairs of noisy and clean ground-truth images to achieve state-of-the-art per-

formance. To alleviate the strict requirement for clean references, self-supervised methods like Noise2Noise [3] loosen this constraint by training denoisers with multiple independent noisy observations of the exact same scene. These two frameworks share a fundamental commonality: *both rely on the availability of multiple independent samples to statistically target the underlying clean signal*; Consequently, their applicability is inherently limited by data-gathering constraints. When only a single noisy image is available—which is the most common yet most challenging scenario—the multi-observation assumption breaks down, making the problem considerably harder. Representative single-image methods, including DIP [7], Self2Self [6], and R2R [5], attempt to overcome this by generating pseudo training pairs from the single available noisy image. This is typically achieved via specialized regularizations or structural manipulation, such as early stopping, Bernoulli sampling with dropout, or re-corruption with additional noise.

Despite recent progress, a critical dilemma remains. While multi-observation methods (supervised and Noise2Noise) are heavily bottlenecked by strict data collection requirements, existing single-image approaches inevitably rely on structural manipulations—such as masking, downsampling, or re-corruption—to construct training targets. These *information-lossy* operations discard vital spatial and texture details, imposing a hard upper bound on denoising quality and leading to visible artifacts, residual noise, or texture loss.

The Positive2Negative (P2N) framework addresses above challenge with a threefold objective: (i) operating strictly on a single noisy image, (ii) avoiding any information-lossy preprocessing, and (iii) preserving the complete information of the original observation throughout training.

The core insight of P2N is that real-world noise is approximately zero-mean and symmetric around the clean signal; thus, the *opposite* of the predicted noise constitutes a plausible noise sample from the same distribution. By adding scaled positive and negative versions of the predicted noise back to the estimated clean image, P2N con-

structs virtual training pairs corresponding to the same underlying clean signal. Enforcing consistency between the denoised outputs of these “positive” and “negative” pairs drives the network to learn robust denoising without discarding any original information. This elegant formulation effectively bypasses the information-loss barrier that has constrained prior self-supervised methods based on a single-image.

2. Related Work

Our prior work focus on the trade-off of Supervised, Self-supervised image denoising, Single Image-Based denoising methods and identify the *information-lossy* barrier that motivates P2N.

2.1. Supervised Denoising

Supervised methods learn a mapping from noisy to clean images using paired training data. The pipeline is straightforward: a clean reference x is corrupted (for example by additive Gaussian noise) to form a noisy input y , and a neural network is trained to regress the clean signal, typically by minimizing $\|f_\theta(y) - x\|$. Although this approach yields excellent performance when sufficient paired data is available [1], the requirement of clean ground-truth images is its central limitation. In real scenes where clean references are difficult or impossible to obtain, the supervised paradigm is simply not applicable.

2.2. Noise2Noise

Noise2Noise [3] was the first widely adopted framework to demonstrate that clean targets are not strictly necessary for training a denoiser. Given two *independent* noisy observations y_1, y_2 of the same scene, the network is trained to map one to the other. Under the assumptions that the noise is zero-mean and the two observations are conditionally independent given the clean image, the optimal least-squares regressor is exactly the clean signal. The practical limitations of Noise2Noise are twofold: (i) obtaining two genuinely independent noisy captures of the same scene is often impractical (any motion or scene change between captures violates the assumption), and (ii) approximating such pairs from a single image — for instance by downsampling or by adding extra synthetic noise — inevitably introduces information loss in the training signal.

2.3. Noise2Void

Noise2Void [2] removes the need for paired noisy images by introducing a *blind-spot* prediction scheme. The network is asked to predict the value of a masked central pixel from its surrounding context only, under the assumptions that (i) noise is pixel-wise independent and (ii) the underlying signal is spatially correlated. While this elegantly enables training on a single noisy image, masking the central pixel

discards a critical piece of evidence at every training position — which is the canonical example of an information-lossy operation. The blind-spot constraint also limits performance on real-world noise that exhibits spatial correlation, since the independence assumption no longer holds.

2.4. Single Image-Based Methods

Several methods specifically target the single-image setting by constructing pseudo training pairs from *one* noisy observation. These methods share the same fundamental issue: constructing pseudo pairs this way inevitably loses information from the original noisy observation, which P2N aims to solve by reusing the predicted noise to construct the training pairs.

The earliest self-supervised single image denoiser Deep Image Prior [7] fits a randomly initialized network to the single noisy image and relies on the network architecture itself as an implicit prior, using early stopping to prevent the network from memorizing the noise.

Self2Self [6] applies Bernoulli sampling (random dropping of input pixels) together with dropout inside the network to generate two distinct stochastic views of the same noisy image. The network is trained to predict the dropped pixels from the remaining ones, and the final prediction is obtained by averaging many stochastic forward passes at inference time. The Bernoulli dropping step, however, removes pixels from the input, introducing information loss in the training pipeline.

Recorrupted-to-Recorrupted (R2R) [5] synthesizes a training pair by adding fresh Gaussian noise to the same noisy image in opposing directions, producing two recorrupted observations that are then mapped to each other. Although R2R operates on a single image, the additional injected noise lowers the effective signal-to-noise ratio of both views, which again amounts to information loss.

3. Methodology

In this section, let us introduce the methodology of Positive2Negative (P2N) which is implemented and evaluated in this project. The P2N paradigm is designed to break the information-lossy barrier prevalent in existing self-supervised single image denoising methods. Current paradigms often rely on downsampling or masking, which inherently lose crucial image information. P2N overcomes this by constructing data that comprehensively covers the original image’s information without discarding pixels.

The foundation of this methodology relies on three key assumptions regarding real-world noise:

- Real-world noise is independent.
- Real-world noise is approximately symmetrical.
- Real-world noise is distributed around a zero mean.

Consequently, the opposite noise, denoted as $-n$, approximately follows the same distribution as the original

noise n . Leveraging this insight, the P2N pipeline utilizes a dual-network approach—a base pre-trained denoiser and a target denoiser—divided into two primary phases: Renoised Data Construction (RDC) and Denoised Consistency Supervision (DCS).

3.1. Renoised Data Construction (RDC)

The goal of RDC is to construct multiple noisy observations from a single image to serve as training data. Because it utilizes the entirety of the image rather than masking it, RDC functions as an information-preserving restoration step rather than purely synthetic data construction.

Given a single dirty image observation $y = x + n$, the process operates as follows:

1. **Initial Denoising (Base Pre-trained denoiser):** The dirty image is passed through a frozen, base pre-trained denoiser (e.g. U-Net) to extract an initial predicted clean image $\hat{x}$.

$$\hat{x} = \mathcal{F}_\theta(y)$$

2. **Noise Extraction:** The predicted noise $\hat{n}$ is isolated by subtracting the predicted clean image from the original noisy observation.

$$\hat{n} = y - \hat{x}$$

3. **Parameter Sampling:** Positive and negative scale parameters, σ_p and σ_n , are sampled from a Gaussian distribution $\mathcal{N}(1, \sigma)$. In our implementation, σ is fixed at 0.75.
4. **Multi-scale Noise Construction:** These scaling parameters are multiplied with the predicted noise to construct multi-scale Positive noise $\hat{n}_p$ and Negative noise $\hat{n}_n$.

$$\hat{n}_p = \sigma_p \odot \hat{n}$$

$$\hat{n}_n = \sigma_n \odot \hat{n}$$

5. **Renoising:** Finally, the constructed noises are applied back to the predicted clean image $\hat{x}$ to generate two new, independent noisy observations, y_p and y_n .

$$y_p = \hat{x} + \hat{n}_p$$

$$y_n = \hat{x} - \hat{n}_n$$

This step successfully generates independent data samples ($y \neq y_p \neq y_n$) from just one single dirty image, setting the stage for self-supervised learning.

3.2. Denoised Consistency Supervision (DCS)

Since both y_p and y_n are constructed from the exact same predicted clean image $\hat{x}$, they inherently correspond to the same underlying clean signal. DCS formulates a supervision strategy based on this observation to fine-tune the second network: the **target denoiser**.

During the training loop (typically around 100 iterations), DCS enforces consistency across the target network’s outputs. Instead of requiring a massive paired dataset, the target denoiser ($\mathcal{F}_\theta$) learns exclusively from the variations of the single dirty image. It supervises the denoiser to ensure that regardless of whether it receives the positive or negative noisy image, the generated output remains the same.

The loss function is defined as:

$$\mathcal{L}_{p2n} = \|\mathcal{F}_\theta(y_p) - \mathcal{F}_\theta(y_n)\|$$

Rather than using a standard Mean Squared Error (MSE) in all cases, the network can employ a gradually varying norm to cover a broader range of unknown noise distributions without requiring parameter fine-tuning:

$$\|x\|_{2 \rightarrow 1.5} = (|x| + \epsilon)^\gamma$$

Where $\epsilon = 10^{-8}$ and γ linearly transitions from 2 to 1.5 during the training process.

By continuously penalizing the difference between the two outputs during these fine-tuning iterations, the target denoiser learns to become completely insensitive to the input noise variations ($\hat{n}_p$ and $\hat{n}_n$). As the loss converges, the fine-tuned model robustly extracts the consistent clean signal $\hat{x}$ directly from the single dirty image.

4. Experiment

4.1. Experimental Setup and Pretraining (Stage 1)

To validate the Positive2Negative (P2N) paradigm, the first stage of our experiment involved establishing a baseline and obtaining a pre-trained model. We implemented a standard Noise2Noise [3] U-Net architecture. Supervised pre-training was conducted using the DIV2K dataset, consisting of 4K resolution images. Synthetic Gaussian noise was added to the training data to simulate noisy real-world conditions.

During this pre-training phase, we evaluated two different pixel intensity scaling ranges for the input data: $[0, 1]$ and $[-1, 1]$. The pre-training phase utilized the following hyperparameters for both configurations:

- **Epochs:** 200
- **Optimizer:** Adam
- **Learning Rate:** 0.001
- **Weight Decay:** 0 (Not used)
- **Beta:** (0.9, 0.99)
- **Batch Size:** 8
- **Crop Size:** 1024×1024
- **Noise Sigma:** 0.1 (ratio)

As shown in Figure 1, the loss converges nicely and the PSNR benchmark continues to rise too. To further validate, initial inference tests were run the SIDD, CC, and

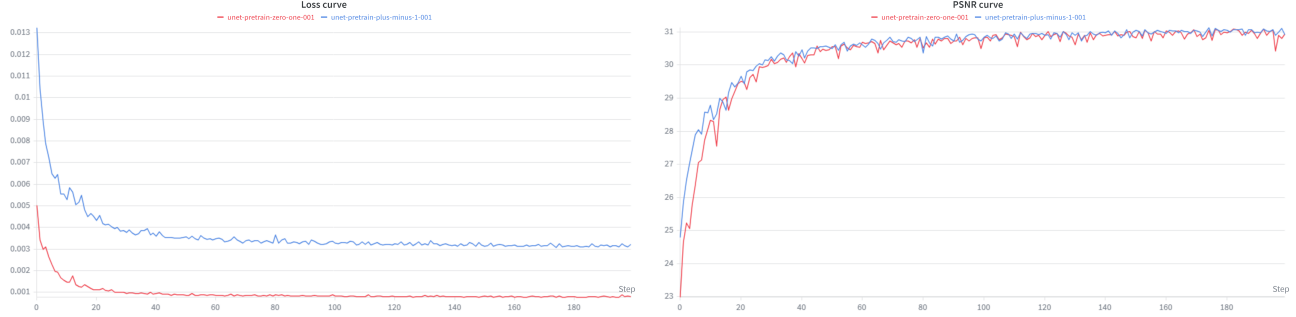


Figure 1. Loss and PSNR curves of the baseline pre-trained U-Net via supervised training with the Gaussian noised DIV2K dataset. The graphs compare the performance of the $[0, 1]$ (red) and $[-1, 1]$ (blue) intensity scaling ranges.

PolyU datasets as shown in Figure 2. This step ensured the base model functioned correctly before applying the self-supervised learning loop.

4.2. P2N Self-Supervised Training Loop (Stage 2)

The core of the experiment focused on implementing the P2N Self-Supervised Learning (SSL) training loop from scratch. We evaluated two distinct initialization strategies for the U-Net prior to the SSL loop even though the paper explicitly said that P2N framework requires a pre-trained denoiser to avoid falling into trivial solutions:

1. **From Scratch:** Initializing the network with random weights.
2. **From Pre-trained:** Initializing the network with the optimal weights obtained from the DIV2K pre-training phase (utilizing the $[0, 1]$ intensity range).

The P2N training loop was executed with the following configuration:

- **Iterations:** 100
- **Learning Rate:** 0.0001
- **Crop Size:** 1024×1024
- **Loss Function Gamma:** $2 \rightarrow 1.5$

As shown in Figure 3, the loss and PSNR curves expose a glaring disconnect that occurs during the self-supervised training phase. On the surface, the loss curves for both initialization methods decrease rapidly and converge near zero, which would normally indicate successful optimization. However, the PSNR curves reveal a catastrophic degradation in actual image quality. The model initialized from scratch starts with an abysmal PSNR that remains flat throughout training. More tellingly, the pre-trained model begins with a strong PSNR—inherited from the baseline—but experiences a steep, continuous decline across all 100 iterations. This quantitative collapse is directly reflected in the visual outputs in Figure 4. The from-scratch model produces a completely black image, while the pre-trained model yields a heavily darkened, blurry result.

These curves perfectly illustrate the network bypassing actual denoising and instead aggressively minimizing the consistency loss by collapsing the image toward a zero-value (black) trivial solution.

4.3. Results and Failure Analysis (Stage 3)

The P2N paper proposes that the model learns to become insensitive to input noise by enforcing consistency between the denoised outputs of a positive noisy image and a negative noisy image. However, our empirical validation revealed a critical instability in this implementation.

During the P2N training loop, the model’s performance collapsed, failing to denoise the images effectively. Visual inspection of the outputs indicated that the images were progressively darkening and losing all structural details.

Mechanism of Failure: The root cause of this failure stems from the model finding a “trivial solution” to minimize the consistency loss. The P2N objective function minimizes the Mean Squared Error (MSE) between the outputs of the two parallel branches. In practice, rather than learning true image priors to separate signal from noise, the network bypassed the intended learning mechanics. It discovered that the easiest way to achieve the smallest MSE was to drive all pixel values toward a uniform color—specifically, by making both images entirely dark. Without sufficient regularization constraints to preserve the original image structure, the network opted for this mathematical shortcut, collapsing the outputs to zero loss.

5. Conclusion

In this study, we implemented and evaluated the Positive2Negative (P2N) self-supervised single-image denoising paradigm to test its theoretical claims of breaking the information-lossy barrier. While the framework presents an elegant approach to generating multi-scale noisy training data without discarding critical spatial information, our empirical validation exposed a severe vulnerability in its practical execution.



Figure 2. Visual comparison of Noised input, Baseline Pre-trained output (using the $[0, 1]$ range model), and Ground Truth.

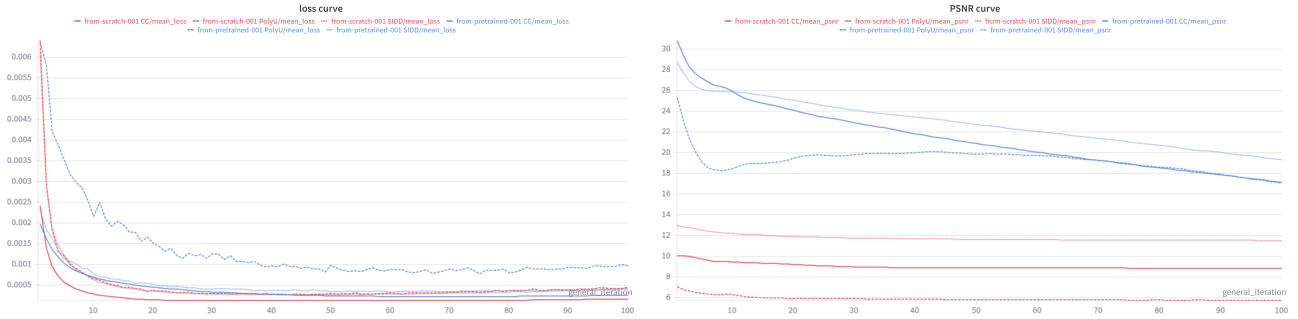


Figure 3. Loss and PSNR curves of the two P2N training methods (initialized from scratch vs. pre-trained baseline).

Despite establishing a highly capable baseline denoiser during the supervised pre-training phase, the P2N consistency supervision loop proved fundamentally unstable. Rather than refining its understanding of natural image priors to isolate noise, the network exploited the unconstrained Mean Squared Error (MSE) objective function. By collapsing the pixel values of both the positive and negative noise branches into a uniform dark state, the model successfully minimized the mathematical loss to near zero while entirely destroying the actual image structure. This resulted in a catastrophic drop in Peak Signal-to-Noise Ratio (PSNR) and the rendering of trivial, black images for both from-scratch and pre-trained initializations.

Ultimately, our experiments demonstrate that the P2N paradigm lacks the necessary guardrails to function reliably. While utilizing symmetrical noise properties is an innovative step forward for single-image self-supervised learning, the architecture requires robust regularization constraints to actively penalize trivial mapping and preserve spatial integrity. Without these structural additions, the model easily falls into mathematical shortcuts, proving that a standalone consistency loss is insufficient for robust real-world denoising tasks.

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Figure 4. Visual comparison on the CC dataset showcasing the Ground Truth, Baseline Pre-trained output, P2N (Pre-trained) output, and P2N (From Scratch) output.

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